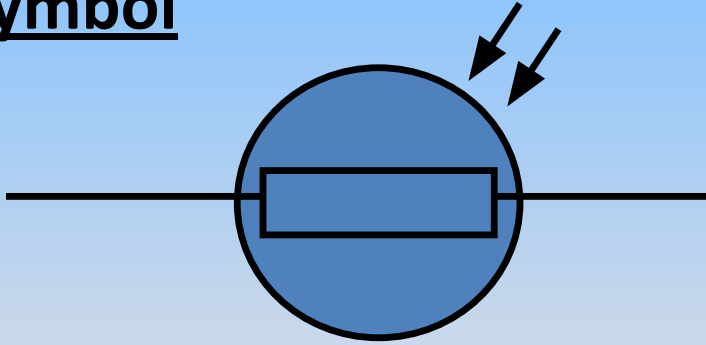


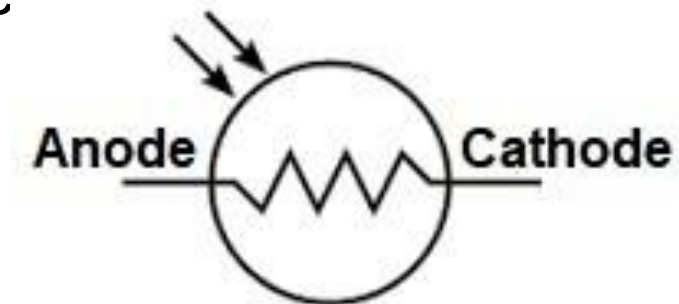
Light Dependent Resistor (L.D.R.)

Circuit symbol



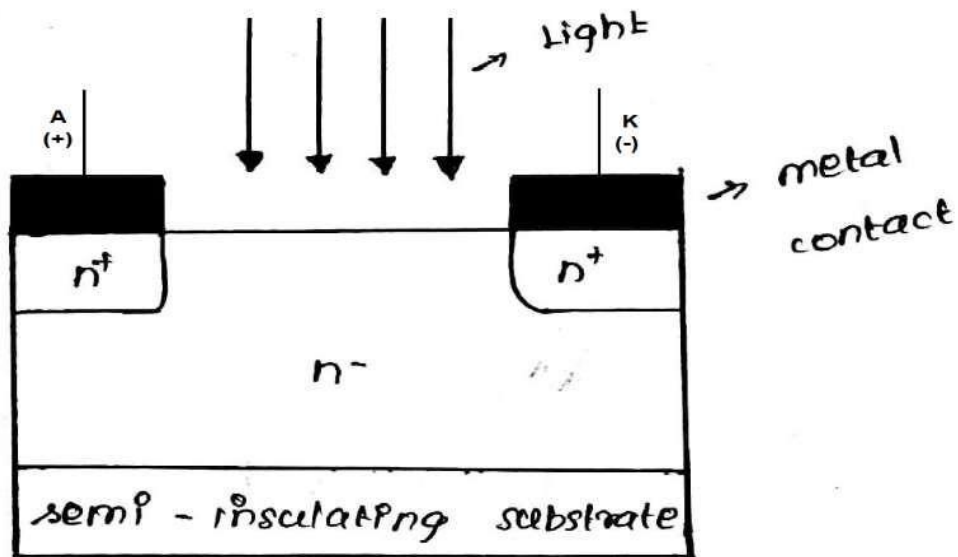
The resistance of the LDR changes as the amount of light (intensity) falling on it changes

A Light Dependent Resistor (LDR) or a photo resistor is a device whose resistivity is a function of the incident electromagnetic radiation. Hence, they are light sensitive devices. They are also called as photo conductors, photo conductive cells or simply photoce^{ll}~



They are made up of semiconductor materials having high resistance. There are many different symbols used to indicate a LDR, one of the most commonly used symbol is shown in the figure below.

The arrow indicates light falling on it.



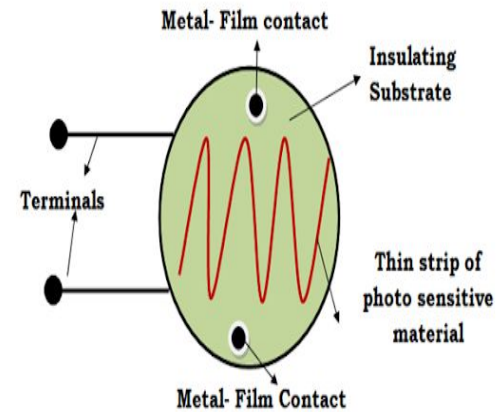
Construction of a Photocell

The structure of a light dependent resistor consists of a light sensitive material which is deposited on an insulating substrate such as ceramic.

The material is deposited in zigzag pattern in order to obtain the desired resistance and power rating. This zigzag area separates the metal deposited are into two regions. Then the ohmic contacts are made on the either sides of the area.

The resistances of these contacts should be as less possible to make sure that the resistance mainly changes due to the effect of light only.

Materials normally used are cadmium **sulphide**, **cadmium selenide**, **indium antimonide** and **cadmium sulphonide**



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Working Principle of LDR

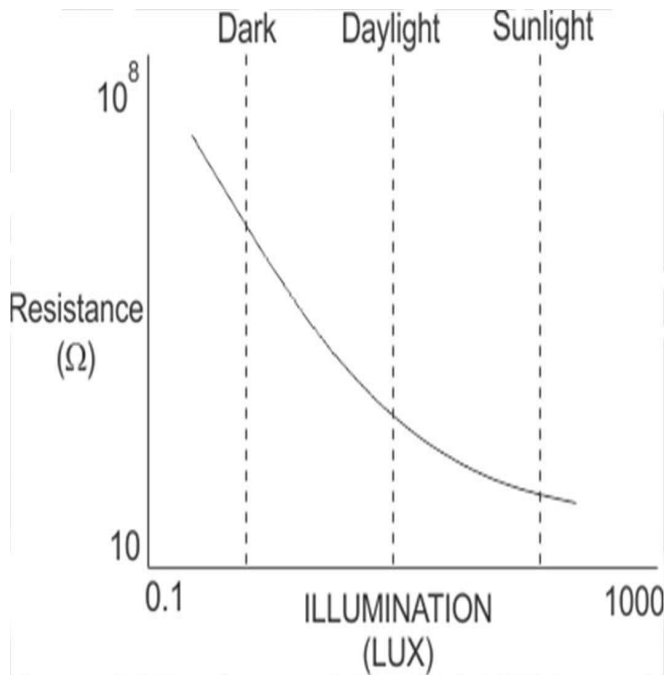
A light dependent resistor works on the principle of **photo conductivity**. Photo conductivity is an optical phenomenon in which the materials conductivity is increased when light is absorbed by the material

when the photons fall on the device, the electrons in the valence band of the semiconductor material are excited to the conduction band.

These photons in the incident light should have energy greater than the band gap of the semiconductor material to make the electrons jump from the valence band to the conduction band.

Hence when light having enough energy strikes on the device, more and more electrons are excited to the conduction band which results in large number of charge carriers.

The result of this process is more and more current starts flowing through the device when the circuit is closed and hence it is said that the **resistance of the device has been decreased**.



Characteristics of LDR

LDR's are light dependent devices whose resistance is decreased when light falls on them and that is increased in the dark.

When a light dependent resistor is kept in dark, its resistance is very high. This resistance is called as **dark resistance**. It can be as high as 1000Ω and if the device is allowed to absorb light its resistance will be decreased drastically. If a constant voltage is applied to it and intensity of light is increased the current starts increasing.

Photocells or LDR's are non linear devices. Their sensitivity varies with the wavelength of light incident on them. Some photocells might not at all respond to a certain range of wavelengths. Based on the **material used different cells have different spectral response curves**

Also, LDR's are less sensitive than photo diodes and phototransistor. (A photo diode and a photocell (LDR) are not the same, a photo-diode is a pn junction semiconductor device that converts light to electricity, whereas a photocell is a passive device, there is no pn junction in this nor it "converts" light to electricity).

Types of Light Dependent Resistors

- **Intrinsic photo resistors** (Un doped semiconductor): These are made of pure semiconductor materials such as silicon or germanium. Electrons get excited from valance band to conduction band when photons of enough energy fall on it and number charge carriers is increased.
- **Extrinsic photo resistors**: These are semiconductor materials doped with impurities which are called as dopants. Theses dopants create new energy bands above the valence band which are filled with electrons. Hence this reduces the band gap and less energy is required in exciting them. Extrinsic photo resistors are generally used for long wavelengths

Applications of LDR

1. Light sensors,
2. Camera light meter,
3. Street lamps, Alarm clock,
4. Burglar alarm circuits,
5. Light intensity meters,
6. Counting the packages moving on a conveyor belt.